



INFRASTRUCTURE

INITIATE. PLAN. ACT.
GLOBAL COMPETENCE IN THERMAL
MANAGEMENT AND SIGNALING
TECHNOLOGY

CONTENTS AT A GLANCE:

TABLE OF CONTENTS



THE PFANNENBERG GROUP:

A GLOBAL NETWORK
TO PROVIDE THE BEST FOR YOU

PAGE 4 – 5



SOLUTIONS FOR BUILDINGS:

SAFELY UNDER COVER

PAGE 6 – 7



AIRPORT SOLUTIONS:

ALWAYS ON THE MARK

PAGE 8 – 9



FULL SPEED AHEAD FOR MORE SECURITY AND SAFETY:

THERMAL MANAGEMENT AND
SIGNALING TECHNOLOGY IN THE
CRANE AND SHIPPING INDUSTRIES

PAGE 10 – 11



WITH ALL THE TRICKS IN THE BOOK:

WATER AND WASTE WATER
SOLUTIONS FROM PFANNENBERG

PAGE 12 – 13



STREET AND TUNNEL SOLUTIONS:

WE CLEAR THE WAY

PAGE 14 – 15



SHARING COMPETENCE:

WE SHARE KNOWLEDGE
TO DUPLICATE IT

PAGE 16 – 17



PFANNENBERG GLOBAL SERVICES:

AT YOUR SIDE WHEN YOU NEED US
WORLDWIDE

PAGE 18 – 19



THE PFANNENBERG GROUP:

A GLOBAL NETWORK TO PROVIDE THE BEST FOR YOU



Pfannenberg is one of the largest globally active companies in the following sectors: of thermal management for electrical enclosures, process cooling and signaling technology. An efficient sales and service network around the globe means we can be there for you quickly and individually around the globe.

To be one of the market leaders, you always have to achieve a little more. Pfannenberg has been doing so for the past 50 years, when Otto Pfannenberg invented the filterfan and thus laid the foundation for thermal management of electrical enclosures.

Today, Pfannenberg offers components, integrated systems and tailor-made application solutions for different tasks in the infrastructure sector such as building services, transportation engineering (tunnel), naval and marine (ships/cranes), airport and also water and waste water in more than 42 countries worldwide.

Furthermore, we are also represented with our products in mechanical engineering, the automotive industry, in medical engineering, by the generation of conventional and renewable energy

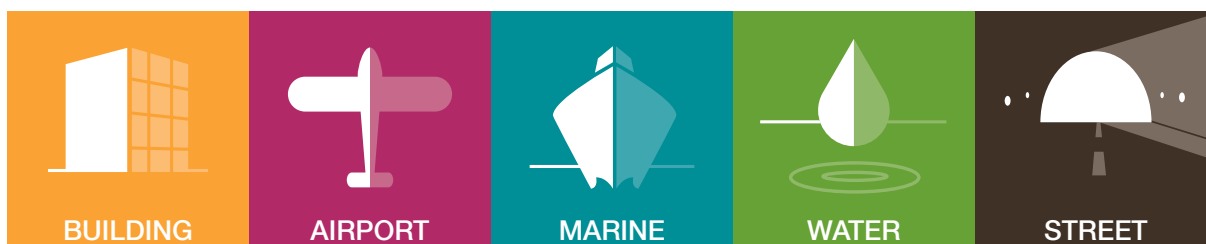
and also in the entire field of manufacturing and processing. Hardly any other company can advise you so comprehensively and supply you from one source like Pfannenberg can.

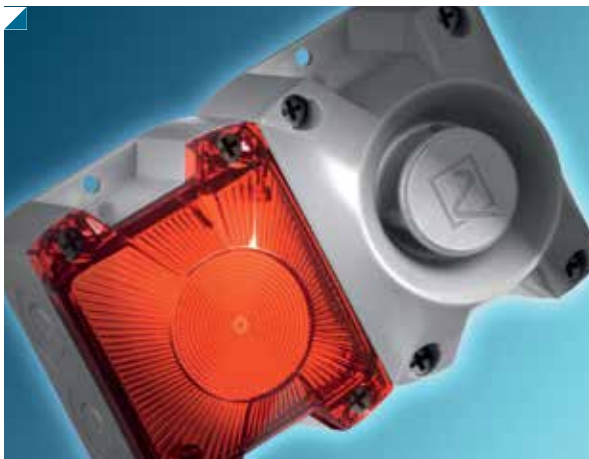
Whether it is the thermal management of electrical enclosures, process cooling, signaling technology, obstacle lights or the worldwide art illuminations by Pfannenberg, which make the Eiffel Tower in Paris amongst other things, sparkle.

As different as our applications may be, they all have one thing in common: they are energy efficient, durable, innovative and in regard to assembly and maintenance, absolutely service oriented.

Regardless of what you are planning, we are always open for requests of all sorts. We will be happy to advise you on combining safety and economic efficiency optimally.

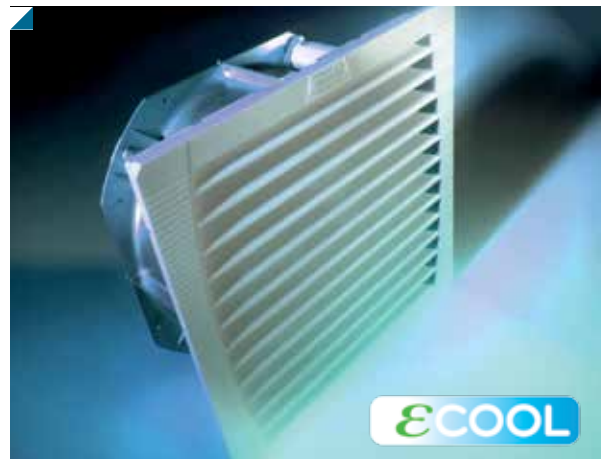
WE HAVE A SOLUTION FOR EVERYTHING:





SIGNALING:

- Visual and Audible Signaling Devices
- Signal Towers
- Ex Signaling Devices
- Art Illumination and Custom Solutions



THERMAL MANAGEMENT:

- Cooling Units
- Air/Water Heat Exchangers
- Chillers
- Air/Air Heat Exchangers
- Filterfans
- Heaters, Thermostats, Hygrostats



10
YEAR WARRANTY
ON SIGNALING
DEVICES!*



SOLUTIONS FOR BUILDINGS:

SAFELY UNDER COVER

Isn't it good to know that we are safe, no matter where we are? Whether we are staying in a hotel, working in an office building or getting on a train at the station, it is our task to create and maintain this safety.

Pfannenberg's solutions make buildings safer – all over the world. Our visible and audible signaling devices alert in the event of danger, fire, burglary, accidents or technical faults. Ranging from gas alarms in garages and rolling gates to obstacle lights on tall buildings. Our individual components and system solutions not only reduce the risk of accidents, however. They also reduce unnecessary downtime and costs and guarantee reliability and consistency.

Whether in houses, schools, public buildings, hospitals or factories: with intelligent thermal management solutions, Pfannenberg reliably and durably protects the building technology and thus in turn protects people. Our filterfans and cooling units provide a perfect climate for the control systems found in lighting, blinds and safety. Heaters and thermostats protect the control systems found in car park barriers, rolling gates, car park ticket machines and access controls from the build-up of condensate. For the use of outdoor displays, the use of air/-air heat exchangers also ensures a hermetic sealing of the enclosures' interiors.



DEFINITELY THE CORRECT SOLUTION FOR ALL INDUSTRIAL AND PUBLIC BUILDINGS:

- Obstacle lights on tall buildings
- Visual alarms for the hearing impaired and for acoustic smog
- Fire and gas alarms
- Thermal management solutions for control equipment

* from the PATROL, PYRA, DS and Quadro series

AIRPORT SOLUTIONS:

ALWAYS ON THE MARK

At hardly any other place are security and safety so much in focus as at the airport. Here, everything has to work smoothly both in front of and behind the scenes. Pfannenberg components and system solutions make an important contribution in this respect.

Pfannenberg has almost become indispensable at many airports. In the tower, **ECOOL** cooling units protect important recording devices from overheating. Air/water heat exchangers cool the X-ray machines in the baggage checking areas.

Pfannenberg heaters, thermostats and hygrostats safely protect the electronics in the car park ticket machines from corrosion. Additionally, signaling devices are used in many different applications.

In contrast to Pfannenberg thermal management

solutions which are often not visible for the visitor, these must always be visible and often audible, so they are noticed in the event of an alarm. Our signaling solutions are used everywhere where safety and a smooth traffic flow are paramount. On towers, buildings and cranes, LED obstacle lights alert the air traffic, in particular airplanes landing and taking off. In the "Baggage Claim" area, continuous lights and flashing lights alert about the start and operation of the conveyor belts. Throughout the airport, signaling devices alert visitors and staff visually and audibly in the event of a fire alarm.

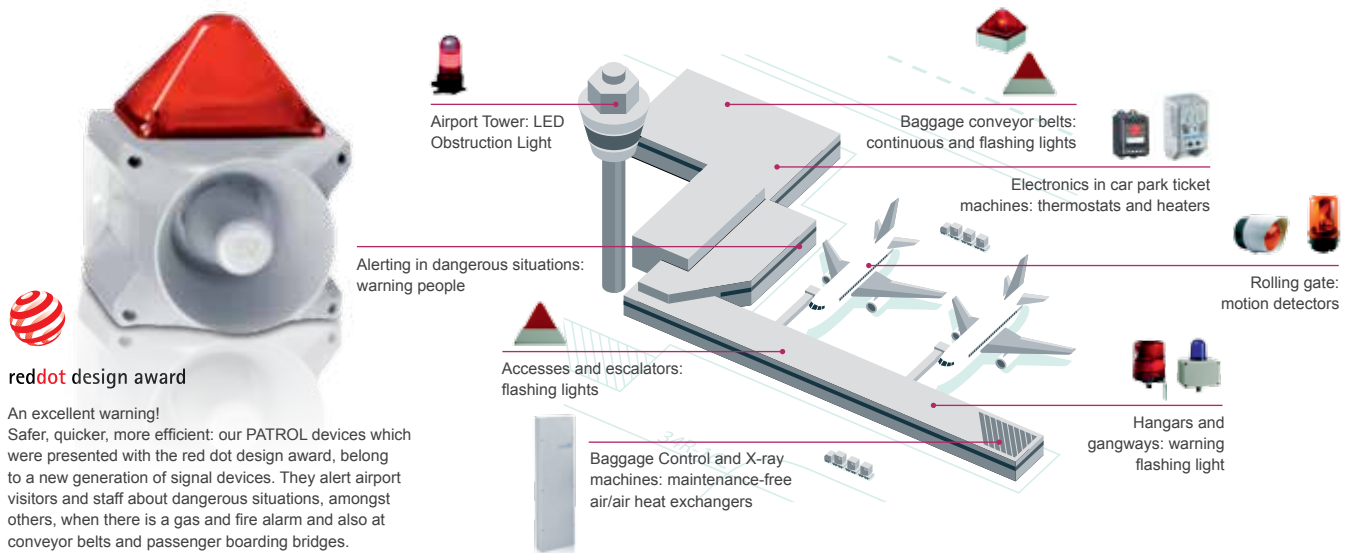


ECOOL COOLING
UNITS REDUCE
ENERGY
CONSUMPTION BY

43%



INDICATION, WARNING, ALARM:



FULL SPEED AHEAD FOR MORE SECURITY AND SAFETY:

THERMAL MANAGEMENT AND SIGNALING TECHNOLOGY IN THE CRANE AND SHIPPING INDUSTRIES

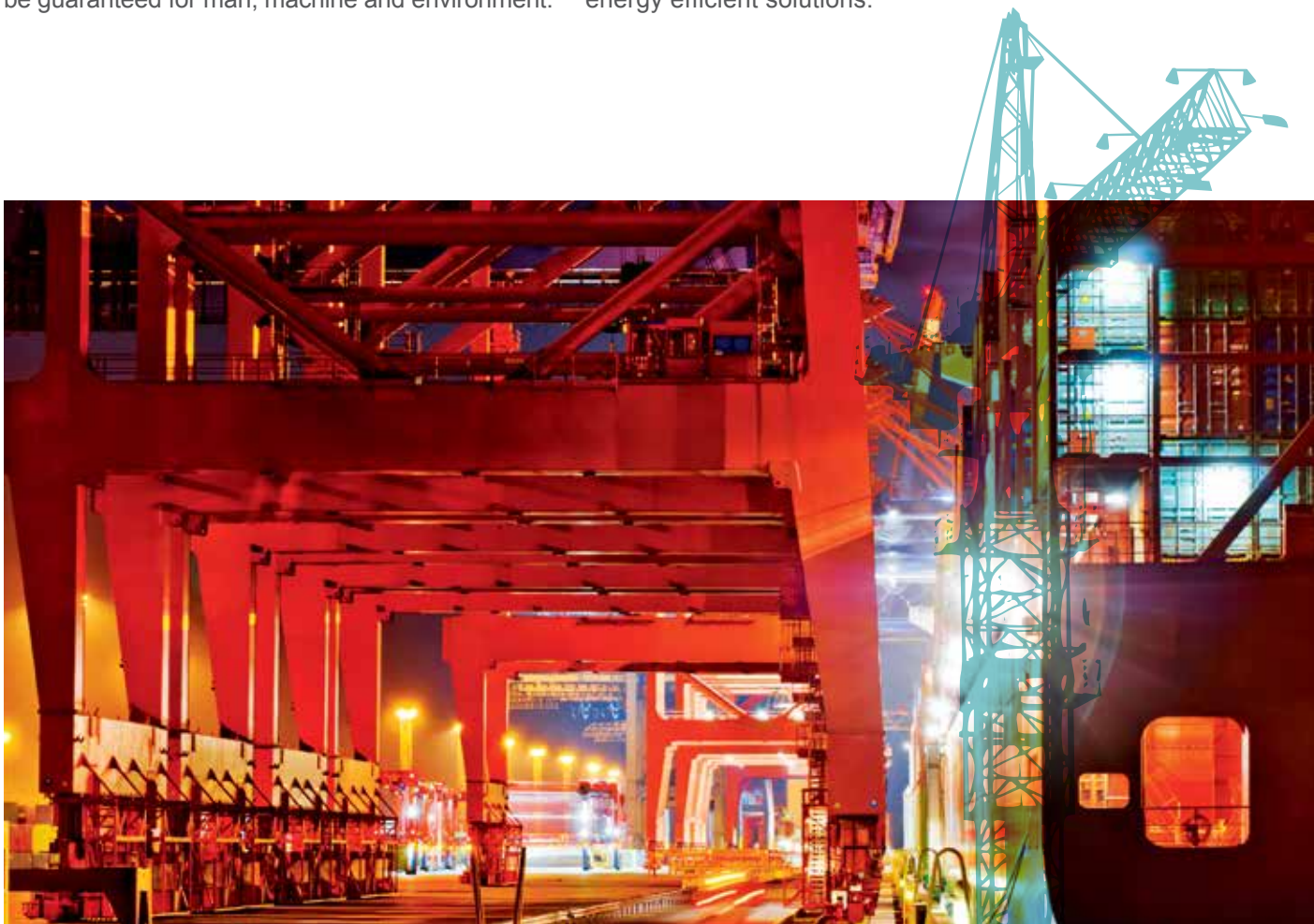
We humans appear tiny in places where freight ships with several thousand containers operate and portal cranes heave capacities of up to 1,000 tonnes. Pfannenberg solutions ensure these forces remain safely under control.

Cranes, elevators and ships are real sloggers in our automated world. Hydraulic cranes, automated traffic guidance systems and gangways have lifted shipping into new spheres of velocity and productivity. Yet, where so much concentrated power is used, it has to be carefully controlled.

Pfannenberg components ensure safety, either separately or as a system solution as well as visually and audibly. Pfannenberg components ensure safety in those areas in which safety must be guaranteed for man, machine and environment.

On land, our applications on cranes and elevators signal loading capacities warn about movements or alert when the wind is too strong.

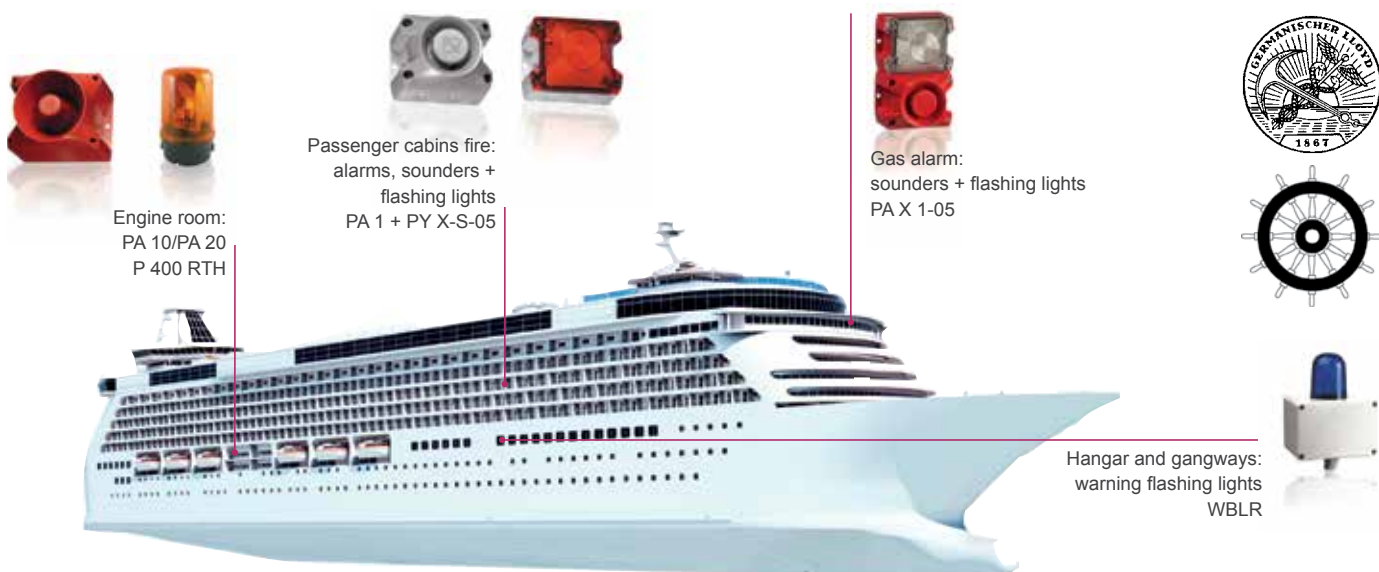
On board ships for example, safety is increased by gas alarms in engine rooms, fire alarms in the cabins and signal lamps on the moving gangways. At the harbour terminals, our lighting solutions guarantee a safe routing of traffic. Filterfans on thermal management devices ensure the controlling devices on cranes are cooled through energy efficient solutions.



24/7

ALL SIGNALING DEVICES
HAVE TO BE IN OPERA-
TION IN HARBOURS FOR
WARNING .

INDICATION, WARNING, ALARM:





100%
INCREASE IN AIR
CAPACITY THANKS TO
PATENTED FANS AND
FILTER TECHNOLOGY

WITH ALL THE TRICKS IN THE BOOK:

WATER AND WASTE WATER SOLUTIONS FROM PFANNENBERG

Drinking water is becoming more valuable all over the world. The careful use of this precious resource is one of man's biggest future tasks. Intelligent and energy efficient solutions for processing and treatment of water and waste water are becoming increasingly more significant. Discover how easy it is to permanently reduce your costs with Pfannenberg and the proper system.

Pfannenberg provides its customers with solutions for every link in the utilization chain – from the exploitation and processing of water to the distribution and sewage treatment. Furthermore, Pfannenberg's signaling devices show the process states of systems and alert in the event of danger, fire, accidents or technical faults.

Whether they are in the ozone systems for processing of water in pumps with variable frequency drive technology or in UV filter

systems, industries all over the world trust our components and solutions in their water carrying systems.

Cooling units from the **ECOOL** range with a top energy efficiency ratio and a long passage of air provide an optimal system solution for cooling of your processes, machines and controls.

The patented 4th Generation filterfans enable an increase in service times of up to 300 % due to their use of patented fluted filters.

VFD pumping systems

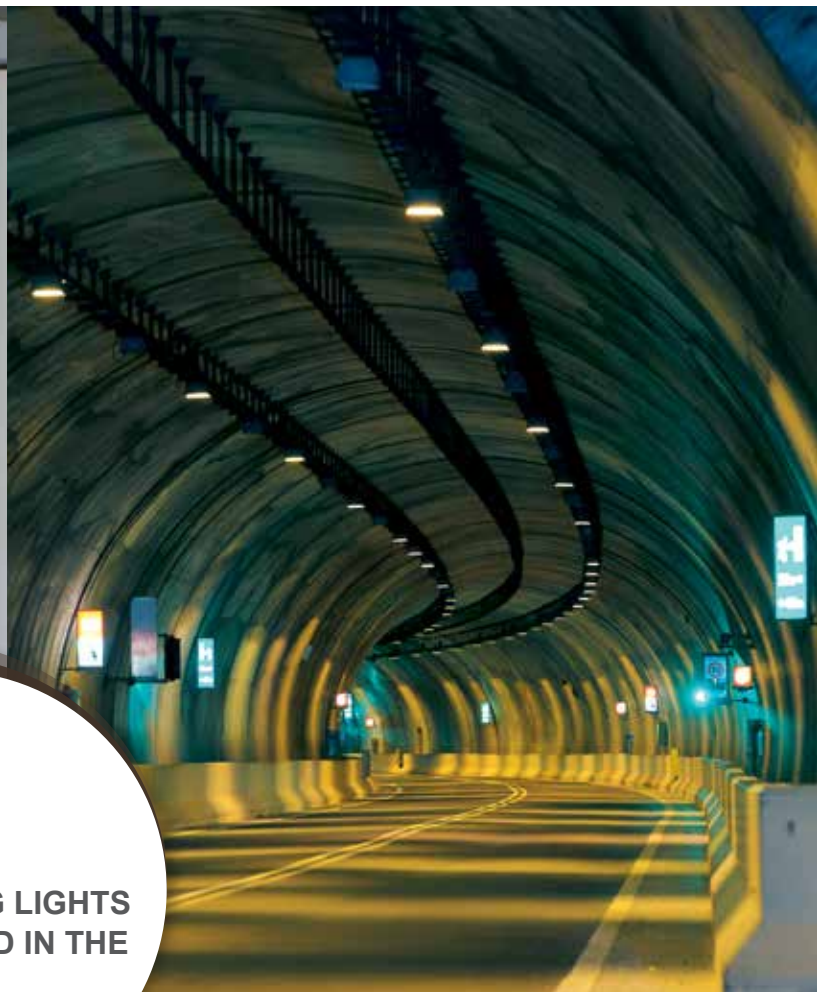


ROAD AND TUNNEL SOLUTIONS:

WE CLEAR THE WAY

The building and safe operation of tunnels not only present a great challenge to engineers. All products and solutions meet the strictest requirements and also need to work perfectly in difficult ambient temperatures. It's great that there is a partner like Pfannenberg who always has a solution when it comes to safety.





1000

**1000 FLASHING LIGHTS
ARE INSTALLED IN THE
PARIS TUNNEL
NETWORK**

Tunnels require special devices and equipment to operate safely. High resistance to pressure change and remote supervision of the operation are just two of the areas in which Pfannenbergs products and system solutions come into use.

Starting with the cooling and condensate protection of machine control systems of the huge tunnel drills, to the fire alarms and to the loud speakers on the emergency escape doors, our solutions are used amongst others, in the Loetschberg tunnel, Gotthard Tunnel, in the tunnel sections of the Paris highways and in Germany's longest tunnel: the almost eight kilometre long Rennsteig Tunnel in Thuringia.

On the roads, traffic guidance systems and environmental measuring devices are subject to extreme temperatures. Pfannenbergs cooling units and heaters, including control devices such

as thermostats and hygrostats protect from the formation of condensate and from overheating and thus enable safe traffic management.



Tunnel safety light:
Quadro S-M Flex



DTT **ecool** top mounted cooling unit



PFH compact heater

85%

OF ALL EMPLOYEES
USE SALESFORCE
FOR KNOW-HOW
TRANSFER

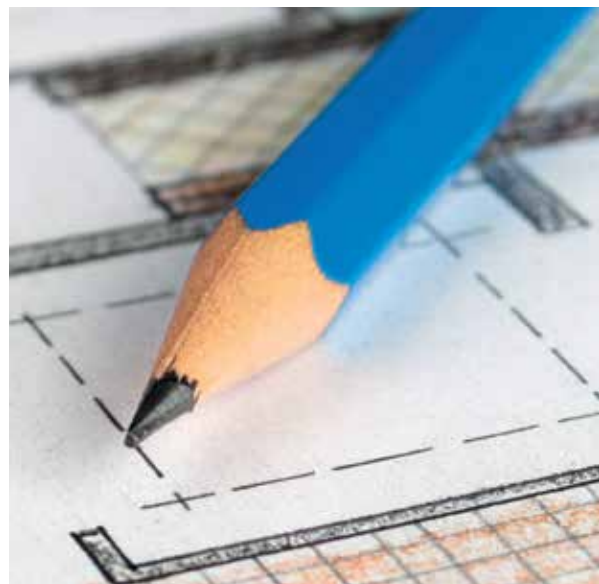
SHARING COMPETENCE:

WE SHARE KNOWLEDGE
TO DUPLICATE IT

The secret of our success: we are always in dialogue with our employees, customers and suppliers. This is the only way we can do justice to the demanding challenges of today and tomorrow, to the high quality demands we have on our products and to our customers.

The first and most important step on the way to a solution at Pfannenberg is to fully understand customers' requirements in order to subsequently design products and services accordingly.

This is possible due to our committed and highly trained employees who, thanks to sound specialist analysis of the tasks, are able to develop solutions that are optimal both technically and cost effectively.





SHARING COMPETENCE FOR YOUR SUCCESS:

- Working group “Operations Excellence for SME”
- Innovation workshops with suppliers and customers
- Symposiums and lectures
- Active support for sales and service partners worldwide
- Planning security by means of standardized products
- Invitations to tender
- Pfannenberg Sizing Software (PSS) for dimensioning



PFANNENBERG GLOBAL SERVICES:

AT YOUR SIDE WHEN YOU NEED US, WORLDWIDE

**You can trust Pfannenberg's products and solutions all over the world.
We have built up a global, comprehensive support system for you to
ensure smooth production at your plants.**

Repair Service

Your cooling system is losing efficiency?
No need to panic: our repair service is there for you immediately – nationally and internationally. Benefit from our worldwide network with a local service without the expensive and cumbersome transportation of your products back and forth.

Spare Parts

If necessary, you can rely on a quick delivery of both wear parts and spare parts. No matter where in the world you need the parts, we deliver original parts as replacements promptly.

Product development

You have new product ideas but lack the resources or experience? Then why not use Pfannenberg's R&D service? Make use of the complete product portfolio that a good

engineering firm has to meet your requirements cost efficiently and concentrate on your own core competence.

Product validation

Need to test a product? You can use Pfannenberg's testing laboratories. Spare the testing capacities in your company and benefit from being able to access professional and comprehensive testing facilities including specialized staff.

Choose your own service level!

In addition to a wide spectrum of services, we offer you the possibility to choose a different service level for every service component in your service and maintenance contract. We will be more than happy to help and advise you with your selection.



Support Organisations – International

Argentina Racklatina Fernando R. Bellora Av. Gaona 2682 Acc. Oeste B 1742KAN, Moreno Phone: +54 0237 405 7000 f.bellora@racklatina.com.ar	Croatia Elektro Partner d.o.o. Slavonska Avenija 24/6 10000 Zagreb Phone: +385 1 618 4793 Telefax: +385 1 618 4795 elektropartner@zg.t-com.hr	India Pfannenberg India 30/53 Kalaimagal Nagar II Main Road Ekkaduthangal Chennai 600 032 Phone: +91 98410 45814 jaya.u@pfannenberg.com.sg	Netherlands ELECTROMACH B.V. Jan Tinbergenstraat 193 7559 SP Hengelo Phone: +31 74 2 472 472 Telefax: +31 74 2 435 925 info@electromach.nl	Spain Pfannenberg Italia s.r.l. Via La Bionda, 13 43036 Fidenza (PR), I Phone: +39 0524 516 711 Telefax: +39 0524 516 790 info@pfannenberg.es
Asia Pfannenberg Asia Pacific Pte Ltd 61 Tai Seng Avenue # B1-01 UE Print Media Hub Singapore 534167 Phone: +65 6293 9040 Telefax: +65 6299 3184 info@pfannenberg.com.sg	Czech Republic Weidmüller, s.r.o. Lomnického 5/1705 140 00 Praha 4 Phone: +420 244 001 400 Telefax: +420 244 001 499 office@weidmueller.cz	Indonesia Guna Elektro GAE Electrical & Mechanical Products Jl. Arjuna Utara 50 Jakarta Barat 11510 Phone: +62 21 565 5010 Telefax: +62 21 568 5030 info@gae.co.id	New Zealand Carrel Electrade Ltd 551 Great South Road Penrose Auckland 1061 Phone: +64 9 525 1753 Telefax: +64 9 525 1756 sales@carrel-electrade.co.nz	Sweden Pfannenberg GmbH Jim Larsen Axel Danielssons väg 271 200 49 Malmö Phone: +46 4037 4876 Mobile: +46 708 878 181 Telefax: +46 4037 4860 jim.larsen@pfannenberg.com
Australia HSC Pfannenberg 21 Rooney Street Maidstone VIC 3012 Phone: +61 4 3550 0712 Telefax: +61 4 3550 0712 susannehaug@hscpfannenberg.com	Denmark Gas Detect Ved Fyret 6 5500 Middelfart Phone: +45 4242 5070 info@gasdetect.dk	Ireland Pfannenberg (UK) Ltd. Unit 6C Aspen Court Bessemer Way Centurion Business Park Rotherham S60 1FB, UK Phone: +44 1709 36 4844 Telefax: +44 1709 36 4211 info@pfannenberg.co.uk	Norway Marin Supply A S Nedre Vei 8, Bykk 155 3183 Horten Phone: +47 3308 3308 Telefax: +47 3308 3309 info@marinsupply.no	Switzerland Carl Geisser AG Hungerbühlstrasse 22 8500 Frauenfeld Phone: +41 44 806 6500 Telefax: +41 44 806 6501 info@carlgeisser.ch
Austria Pfannenberg GmbH Manfred Hartner Grubweg 17 D 8580 Köflach Mobile: +43 664 245 1333 Telefax: +43 3144 6581 manfred.hartner@pfannenberg.com	Estonia Autosafe OY Uranuksenkuja 10 01480 Vantaa, FIN Phone: +358 9 2709 0120 Telefax: +358 9 2709 0129 autosafe@autosafe.fi	Israel ATEKA LTD. Simtat Ha Tavor 4 Industrial Area Segula Petach Tikva 49691 Phone: +972 073 200 1311 Telefax: +972 3 924 3273 marketing@ateka.co.il	Philippines GSPECS Industrial Corporation 179-W 24th avenue, East Rembo Makati City, Philippines 1216 Phone/Telefax: +632 738 0328 sales@gspecs.com.ph	Thailand ND Electric Company 338/139 Soi Lat Phrao 80 Lat Phrao Rd., Wang Thong Lang Bangkok 10310 Phone: +66 2539 6430 Telefax: +66 2539 6655 info@ndelectric.co.th
Belgium I E x T n.v. Heiveldkens 8 - 2550 Kontich Phone: +32 3 458 2741 Telefax: +32 3 458 2761 info@iext.be	Finland Autosafe OY Uranuksenkuja 10 01480 Vantaa Phone: +358 9 2709 0120 Telefax: +358 9 2709 0129 autosafe@autosafe.fi	Italy Pfannenberg Italia s.r.l. Via La Bionda, 13 43036 Fidenza (PR) Phone: +39 0524 516 711 Telefax: +39 0524 516 790 info@pfannenberg.it	Poland Pfannenberg GmbH Andrzej Kushka Phone: +48 667 414 147 andrzej.kushka@pfannenberg.com Olga Kozłowska Phone: +48 665 414 199 olga.kozłowska@pfannenberg.com	Turkey Endaks Endustriyel Aksesuarlar LDT.STI. Perpa Ticaret Merkezi A Blok Kat 5 No. 292 34384 Okmeydanı - Istanbul Phone: +90 212 222 2275 Telefax: +90 212 220 1047 info@endaks.com
Brazil Pfannenberg do Brasil Indústria e Comércio Ltda. Rod. SP-73, 4509 NR Condomínio - Galpão 11 Indaiatuba, SP - 13347-390 Phone/Telefax: +55 19 3935 7187 info@pfannenbergdobrasil.com.br	France AE & T Applications Electroniques & Techniques 4, Impasse Joliot Curie - BP 25 64110 Jurançon Phone: +33 5 5906 0600 Telefax: +33 5 5906 4463 info@aet.fr	Kazakhstan Electric Light Auezova str. 84, office 310 050008 Almaty Phone: +7 727 245 3535 Telefax: +7 727 245 3581 wgm@wgm.kz	Portugal Pfannenberg Italia s.r.l. Via La Bionda, 13 43036 Fidenza (PR), I Phone: +39 0524 516 711 Telefax: +39 0524 516 790 info@pfannenberg.es	Ukraine TEKO INTERFACE TOB 1) Ul. Urliuzkogo 13 09100 Bila Zerkwa Phone: +38 04563 46580 Telefax: +38 04563 46581 2) Ul. Lebanewskogo 6 03058 Kiev Phone: +38 044 374 0640 Telefax: +38 044 374 0642
Bulgaria Eurotrade X Ltd. 176, Brezovsko shossé Blvd. „Dekom Center“ 4003 Plovdiv Phone: +359 3223 5023 Telefax: +359 3223 5022 office@eurotrade-x.com	Greece Pfannenberg Italia s.r.l. Via La Bionda, 13 43036 Fidenza (PR), I Phone: +39 0524 516 711 Telefax: +39 0524 516 790 info@pfannenberg.it	Korea Pfannenberg Asia Pacific Pte Ltd 61 Tai Seng Avenue # B1-01 UE Print Media Hub Singapore 534167 Phone: +65 6293 9040 Telefax: +65 6299 3184 info@pfannenberg.com.sg	Romania R.T.S. Electro 11, Petru Rares Street 011101 Bucharest 1 Phone: +40 21 260 1021 Telefax: +40 21 222 3097 office@rtselectro.ro	United Kingdom Pfannenberg (UK) Ltd. Unit 6C Aspen Court Bessemer Way Centurion Business Park Rotherham S60 1FB Phone: +44 1709 36 4844 Telefax: +44 1709 36 4211 info@pfannenberg.co.uk
Canada Pfannenberg Inc. 68 Ward Road Lancaster, N.Y. 14086, USA Phone: +1 716 685 6866 Telefax: +1 716 681 1521 info@pfannenbergusa.com	Gulf Region Golden Sands Trading Co LLC Level 35, Citadel Tower Business Bay Dubai Phone: +971 4 457 2122 Telefax: +971 4 457 2144 vasu2000@emirates.net.ae	Malaysia A-Comk Techniques Sdn Bhd. 1257-0-6 Capitol Industrial Centre, Batu 6-3/4, Jalan Sungai Besi 57100 Kuala Lumpur Phone: +603 9056 3502 Telefax: +603 9056 3504 sales@acomk.com	Slovakia Elektris s.r.o. Elektrarenska 1 831 04 Bratislava Phone: +421 2 4920 0111 Telefax: +421 2 4920 0199 bratislava@elektris.sk	United States of America Pfannenberg Inc. 68 Ward Road Lancaster, N.Y. 14086 Phone: +1 716 685 6866 Telefax: +1 716 681 1521 info@pfannenbergusa.com
Chile NDU INGENIERIA LTDA. Santiago. 18 de Septiembre 0137 Comuna El Bosque Phone: +56 600 401 1004 Telefax: +56 02 526 5046 info@ndu.cl	Honduras Cilasa Angel Mena Barrio Los Andes 7 Calle, 14 Y15 Ave. N.O. San Pedro Sula Phone: +504 557 1146 angel.mena@leclilasa.com	Mauritius Mubelo Electrical Ltd Office 26, Gateway building St. Jean Road Quatre Bornes Phone: +230 4670 989 Telefax: +230 465 4051 mubelo.electrical@orange.mu	Slovenia Elektrospoji d.o.o. Stegne 25 1000 Ljubljana Phone: +386 1 511 3810 Telefax: +386 1 511 1604 info@elektrospoji.si	Venezuela Somerina, C.A. Qta Corazon de Jesus 4ta Transversal c/c El Carmen. Los Dos Caminos Caracas 1070 A Phone: +58 212 235 1081 Telefax: +58 212 239 9341 klocmoeller@cantv.net
China Pfannenberg (Suzhou) Pte Ltd First Floor, Unit D, Block 5 Modern Industrial Park No. 333 Xingpu Rd. Suzhou Industrial Zone Suzhou 215021, Jiangsu Province Phone: +86 512 6287 1078 Telefax: +86 512 6287 1077 info@pfannenberg.cn	Hungary Weidmüller Kft. Gubacsi út 6 1097 Budapest Phone: +36 1 382 7700 Telefax: +36 1 382 7701 info@weidmueller.hu	Mexico Latin America Electrical Representatives Hermosillo 95 J. de Morelos 5a secc. Ecatepec, Edo de Mexico, CP 55070 Phone: +55 5837 7716 bcontreras@lareps.com	South Africa Phambili Interface (Pty) Ltd 5 Bundo Road, Sebenza Edenvale, P.O. Box 193 Edenvale, 1610 Phone: +27 11 452 1930 Telefax: +27 11 452 6455 alockyer@weidmuller.co.za	
Colombia INGEPRO LTDA. Compania Importadora Comercial Av. Eldorado No. 84A-55 Local 118 A.A. 95406 Santa Fe de Bogotá D.C. Phone: +57 1 410 2621 Telefax: +57 1 295 2581				

PFANNENBERG GROUP, WITH OUR SUBSIDIARIES.

Pfannenberg GmbH
Werner-Witt-Straße 1
21035 Hamburg
Germany

Phone: +49 40 73412 156
Fax: +49 40 73412 101
Email: customercare@pfannenberg.com
Web: www.pfannenberg.com

Pfannenberg United Kingdom, Rotherham Phone:
+44 1709 36 4844
Email: info@pfannenberg.co.uk

Pfannenberg Italia, Fidenza (PR)
Phone: +39 0524 516 711
Email: info@pfannenberg.it

Pfannenberg USA, N.Y.
Phone: +1 716 685 6866
Email: info@pfannenbergusa.com

Pfannenberg Singapore, Singapore
Phone: +65 6741 3562
Email: info@pfannenberg.com.sg

Pfannenberg China, Suzhou
Phone: +86 512 6287 1078
Email: info@pfannenberg.cn

Pfannenberg Brazil, Indaiatuba
Phone: +55 19 3935 7187
Email: info@pfannenbergdobrasil.com.br

Deliveries are made on the basis of the General Terms and Services
of the ZVEI. Subject to technical amendments and misprints.
This paper has been manufactured from chlorine-free bleached cellulose.

